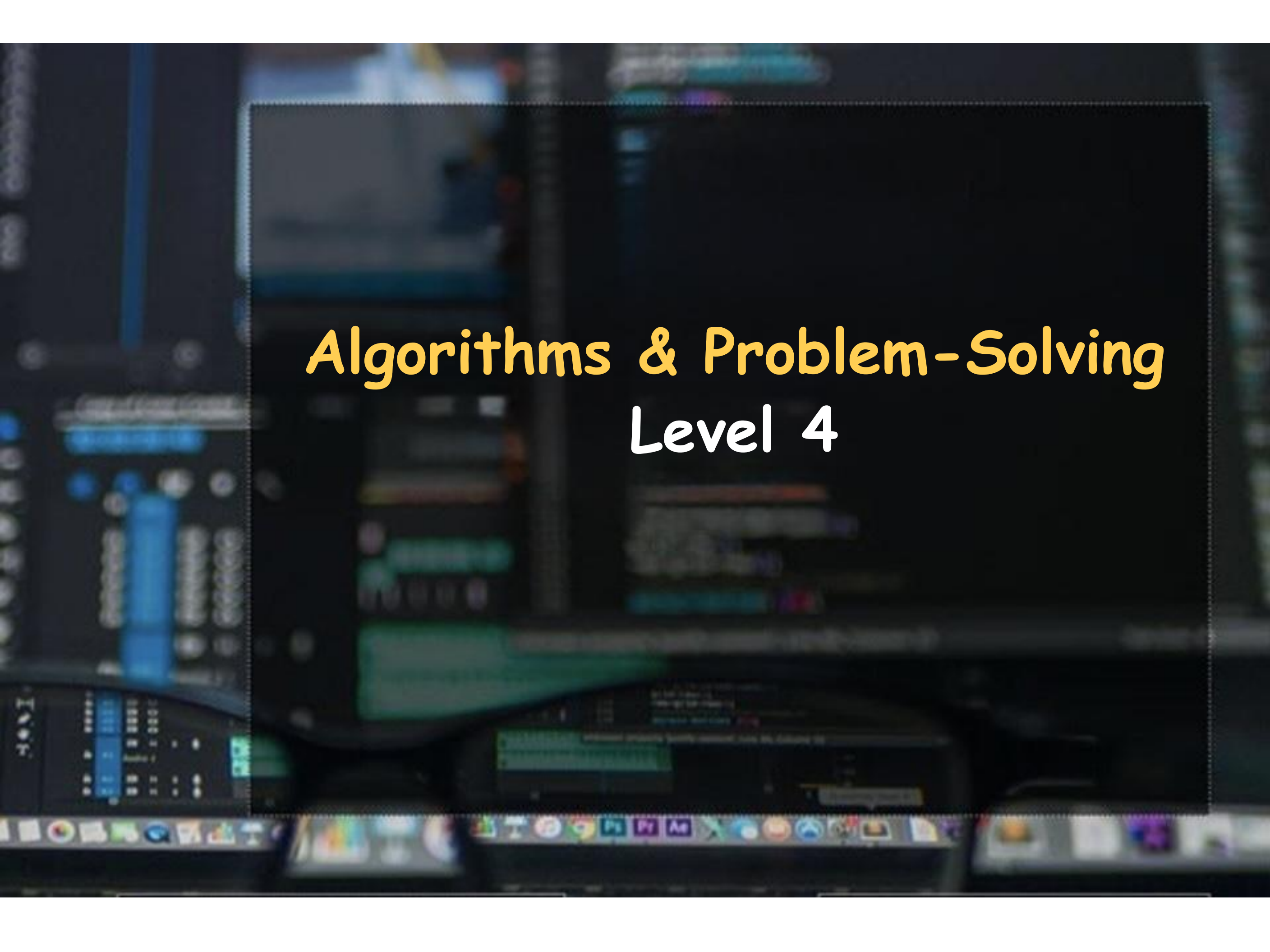
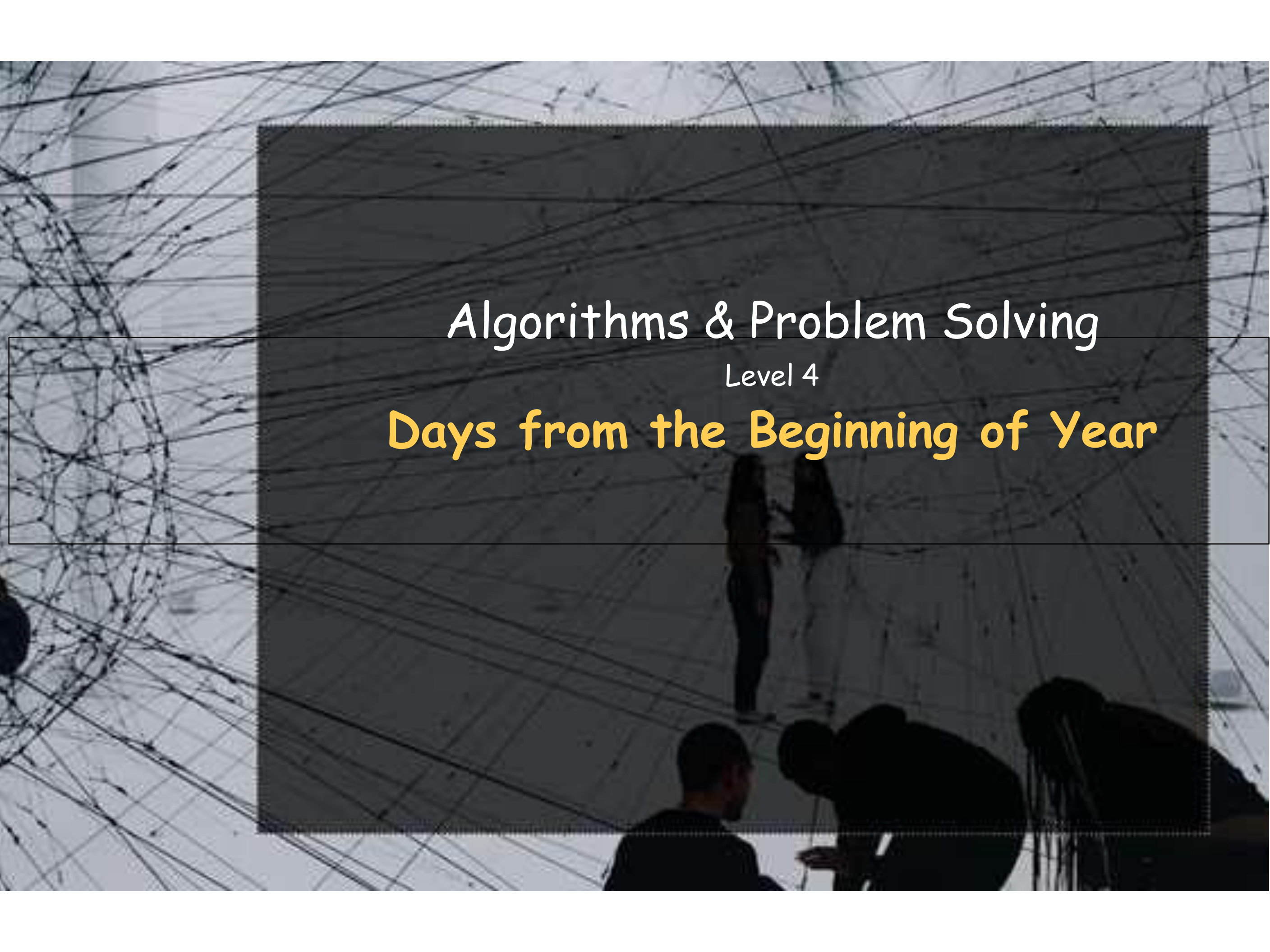




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

Days from the Beginning of Year

Definition

-  The goal is a program to calculate the number of days from the beginning of the year until a specific date entered by the user (day, month, year)

Explanation:

- The program relies on knowing the number of days in each month.
- It adds up the number of days in all months preceding the desired month, then adds the number of days in the current month.
- It takes into account leap years to adjust the number of days in February to 29 instead of 28.

Procedures

1 Read values from the user

-  ReadYear() Enter the year
-  ReadMonth() Enter the month
-  ReadDay() Enter the day

2 Checking for Leap Years

-  Using the `bool` function `IsLeapYear(int year)`
-  Checks if a year is a leap year using the condition:
 $(\text{year} \% 4 == 0 \ \&\& \ \text{year} \% 100 != 0) \ || \ (\text{year} \% 400 == 0)$

3 Calculating the number of days in a month

-  Using `int NumberOfDaysInAMonth(int month, int year)`
-  Returns 29 for February if it's a leap year

4 Calculate the number of days since the beginning of the year

-  By adding the days of the previous months to the current day:
``int NumberOfDaysFromTheBeginningOfTheYear(int day, int month, int year)``



Example:

Please enter a Day? 20

Please enter a Month? 5

Please enter a Year? 1981

Number of Days from the beginning of the year is 140



Note:

- The calculation assumes that the previous months have correct values.
- If a month outside the range (1-12) is entered, the function returns zero.

Summary

- ✓ The program adds up the days of previous months to show how many days have passed since the beginning of the year until the specified date.
- ✓ It accurately takes leap years into account.

Thank you 😊